Reflection and Refraction

Introduction to the laws of reflection and refraction in linear dielectrics

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1 Maxwell's equations and the boundary conditions

Let's start with the Maxwell's equations:

Axiom 1.1 (Maxwell's equations).

The four fundamental equations for the electromagnetic field are

$$\iint_{\partial V} \mathbf{D}\cdot \mathrm{d}\mathbf{A} = \iiint_{V} \rho_f \,\mathrm{d}V \qquad \text{(A1)} \qquad \oint_{\partial S} \mathbf{E}\cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\iint_{S} \mathbf{B}\cdot \mathrm{d}\mathbf{A} \qquad \text{(A2)}$$

$$\iint_{\partial V} \mathbf{B}\cdot \mathrm{d}\mathbf{A} = 0 \qquad \text{(A3)} \qquad \oint_{\partial S} \mathbf{H}\cdot \mathrm{d}\mathbf{l} = \iint_{S} \mathbf{J}_f\cdot \mathrm{d}\mathbf{A} + \frac{\mathrm{d}}{\mathrm{d}t}\iint_{S} \mathbf{D}\cdot \mathrm{d}\mathbf{A} \qquad \text{(A4)}$$

The constitutive relations are

$$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P} \qquad \mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}.$$

1.1 Boundary conditions at a medium-discontinuity

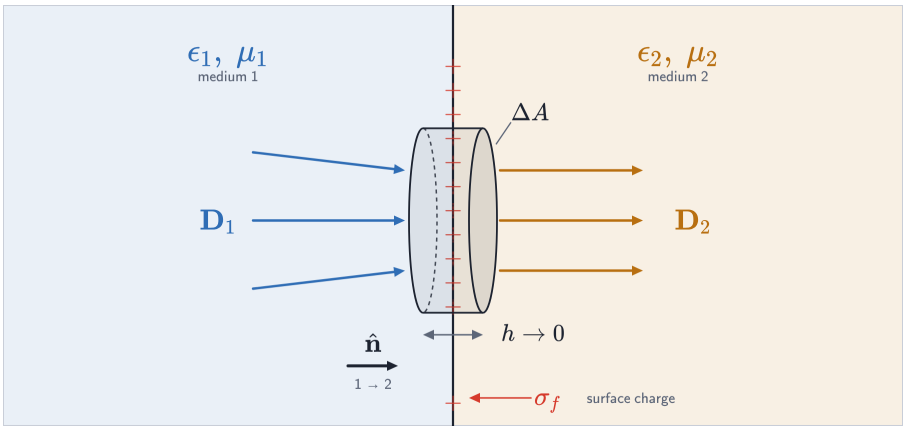


Figure 1: The Gaussian pillpox

1. As illustrated in the figure, consider a Gaussian pillbox of cross-sectional area A and height h at the interface between media 1 and 2. Let $\mathbf{n}$ be the unit normal directed from medium 1 to medium 2. By Gauss's law for the electric displacement field,

$$\oint_{\partial V} \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}} = \sigma_f A,$$

thus,

$$\mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f.$$

2. Applying Gauss's law for the magnetic flux density to the same pillbox gives

$$\oint_{\partial V} \mathbf{B} \cdot d\mathbf{a} = 0 \quad \Longrightarrow \quad \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0.$$

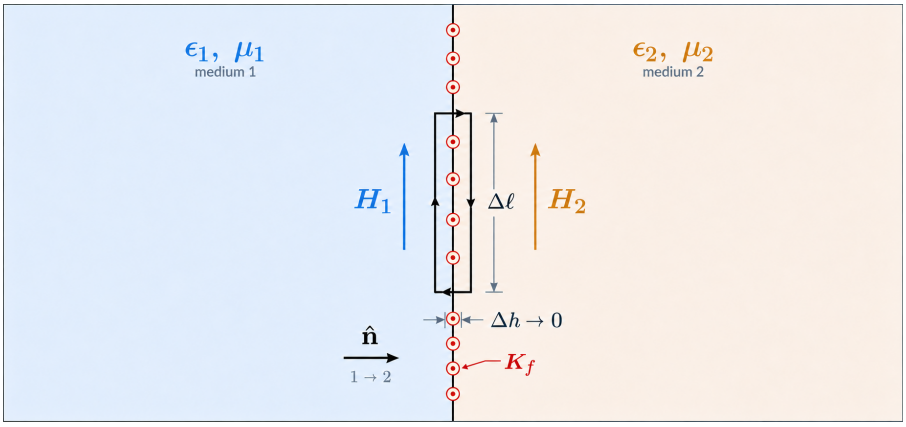


Figure 2: The amperian loops

1. Consider an infinitesimal rectangular loop on the interface. We have:

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

As the loop height tends to zero, the magnetic flux through its area vanishes. So the tangential component of the electric field is continuous:

$$\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}.$$

2. Applying the Ampère–Maxwell law to the same loop gives

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I_{f,\text{enc}} + \frac{d}{dt} \int_S \mathbf{D} \cdot d\mathbf{a}.$$

As the loop height tends to zero, the displacement-flux term vanishes:

$$\mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}_{s,f}.$$

Theorem 1.2 (Boundary conditions for discontinuous media).

Let $\mathbf{n}$ be the unit normal directed from medium 1 to medium 2. The boundary conditions are

$$\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}, \qquad \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}_f,$$
$$\mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f, \qquad \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0.$$

Corollary 1.3 (Boundary conditions for linear isotropic media).

For linear, isotropic media,

$$\mathbf{D} = \varepsilon \mathbf{E}, \qquad \mathbf{B} = \mu \mathbf{H}.$$

Therefore, at the interface,

$$\mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}, \qquad \mathbf{n} \times \left(\frac{\mathbf{B}_2}{\mu_2} - \frac{\mathbf{B}_1}{\mu_1} \right) = \mathbf{K}_f,$$
$$\mathbf{n} \cdot (\varepsilon_2 \mathbf{E}_2 - \varepsilon_1 \mathbf{E}_1) = \sigma_f, \qquad \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0.$$

If there is no free surface current or surface charge density, the boundary conditions reduce to

$$E_{2\parallel} = E_{1\parallel}, \qquad \frac{B_{2\parallel}}{\mu_2} = \frac{B_{1\parallel}}{\mu_1},$$
$$\varepsilon_2 E_{2\perp} = \varepsilon_1 E_{1\perp}, \qquad B_{2\perp} = B_{1\perp}.$$

2 Plane Waves

Consider a plane wave of the form

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}.$$

From Faraday's law we could relate the electric and the magnetic field as:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \implies \mathbf{B} = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}.$$

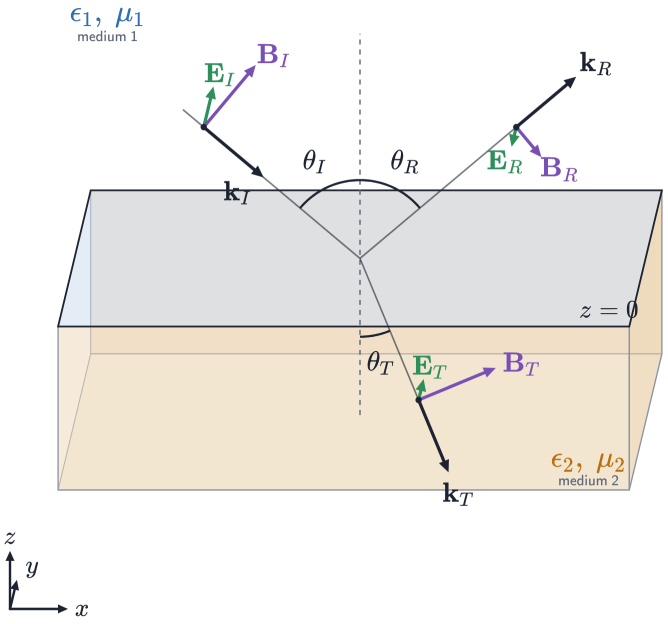


Figure 3: A plane wave incident on a surface (the xy plane). Note that the wave is s-polarised, i.e. the electric field is parallel to the surface of incidence.

Let the wave vectors for the incident, reflected, and transmitted waves be

$$\mathbf{k}_i = k_1 \left(\sin\theta_i\, \hat{\mathbf{x}} - \cos\theta_i\, \hat{\mathbf{z}}\right) \quad \mathbf{k}_r = k_1 \left(\sin\theta_r\, \hat{\mathbf{x}} + \cos\theta_r\, \hat{\mathbf{z}}\right), \tag{medium 1}$$

$$\mathbf{k}_t = k_2 \left(\sin\theta_t\, \hat{\mathbf{x}} - \cos\theta_t\, \hat{\mathbf{z}}\right). \tag{medium 2}$$

Assume the electric fields are polarized along the y-axis:

$$\mathbf{E}_i(\mathbf{r},t) = E_0 \hat{\mathbf{y}}\, e^{i(\mathbf{k}_i\cdot\mathbf{r}-\omega t)} \quad \mathbf{E}_r(\mathbf{r},t) = E_{r0} \hat{\mathbf{y}}\, e^{i(\mathbf{k}_r\cdot\mathbf{r}-\omega t)}, \tag{medium 1}$$

$$\mathbf{E}_t(\mathbf{r},t) = E_{t0} \hat{\mathbf{y}}\, e^{i(\mathbf{k}_t\cdot\mathbf{r}-\omega t)}. \tag{medium 2}$$

Using the relation between the magnetic field and the electric field:

$$\mathbf{B}_i(\mathbf{r},t) = \frac{k_1 E_0}{\omega} \left(\cos\theta_i\, \hat{\mathbf{x}} + \sin\theta_i\, \hat{\mathbf{z}}\right) e^{i(\mathbf{k}_i\cdot\mathbf{r}-\omega t)} \quad \mathbf{B}_r(\mathbf{r},t) = \frac{k_1 E_{r0}}{\omega} \left(-\cos\theta_r\, \hat{\mathbf{x}} + \sin\theta_r\, \hat{\mathbf{z}}\right) e^{i(\mathbf{k}_r\cdot\mathbf{r}-\omega t)} \tag{medium 1}$$

$$\mathbf{B}_t(\mathbf{r},t) = \frac{k_2 E_{t0}}{\omega} \left(\cos\theta_t\, \hat{\mathbf{x}} + \sin\theta_t\, \hat{\mathbf{z}}\right) e^{i(\mathbf{k}_t\cdot\mathbf{r}-\omega t)}. \tag{medium 2}$$

2.1 Laws of reflection andd refraction

At the interface $z = 0$, continuity of the tangential electric field gives

$$E_{i\parallel} + E_{r\parallel} = E_{t\parallel}.$$

Since all three electric fields are polarized along the y-axis, this becomes

$$E_0 e^{i(k_1 \sin\theta_i\, x - \omega t)} + E_{r0} e^{i(k_1 \sin\theta_r\, x - \omega t)} = E_{t0} e^{i(k_2 \sin\theta_t\, x - \omega t)}.$$

1. Since the boundary condition must hold for every x, and all three waves have the same frequency, their tangential phase components must match:

   $$k_1 \sin\theta_i = k_1 \sin\theta_r = k_2 \sin\theta_t.$$

   The first equality gives

   $$\theta_i = \theta_r,$$

   which is the law of reflection.

2. The second equality gives

   $$k_1 \sin\theta_i = k_2 \sin\theta_t.$$

   For a plane wave in a homogeneous isotropic medium,

   $$k_j = \frac{n_j \omega}{c}.$$

   where n_j is the refractive index. Since the frequency is the same in both media, this becomes

   $$n_1 \sin\theta_i = n_2 \sin\theta_t,$$

   which is Snell's law.

Theorem 2.1 (Law of reflection and Snell's law).

1. According to the law of reflection, if θ_i and θ_r are the angles of incidence and reflection, respectively, then

   $$\theta_i = \theta_r.$$

2. For an interface between two media with refractive indices n_1 and n_2, Snell's law is

   $$n_1 \sin\theta_i = n_2 \sin\theta_t,$$

   where θ_i and θ_t are the angles of incidence and refraction, respectively.

2.2 Fresnel's equations

At $z = 0$, continuity of the tangential electric field gives

$$E_0 + E_{r0} = E_{t0}.$$

Continuity of the tangential magnetic field gives

$$\frac{k_1 \cos\theta_i}{\omega\mu_1} E_0 - \frac{k_1 \cos\theta_r}{\omega\mu_1} E_{r0} = \frac{k_2 \cos\theta_t}{\omega\mu_2} E_{t0}.$$

Using $\theta_r = \theta_i$, this becomes

$$\frac{k_1 \cos\theta_i}{\omega\mu_1}(E_0 - E_{r0}) = \frac{k_2 \cos\theta_t}{\omega\mu_2} E_{t0}.$$

Define the reflection and transmission coefficient for the electric-field amplitudes as

$$R_s = \frac{E_{r0}}{E_0}, \qquad T_s = \frac{E_{t0}}{E_0}.$$

The intrinsic impedance in medium j is

$$\eta_j = \sqrt{\frac{\mu_j}{\varepsilon_j}}, \qquad \frac{k_j}{\omega\mu_j} = \frac{1}{\eta_j}.$$

Therefore, the two boundary equations become

$$1 + R_s = T_s, \qquad \frac{\cos\theta_i}{\eta_1}(1 - R_s) = \frac{\cos\theta_t}{\eta_2} T_s.$$

Solving for the reflection coefficient and transmission coefficient gives

$$R_s = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}, \qquad T_s = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}.$$

Using Snell's law, we can write the reflection coefficient and transmission coefficient in terms of θ_i alone. From Snell's law, we eliminate θ_t using

$$\cos\theta_t = \sqrt{1 - \sin^2\theta_t} = \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}.$$

Substituting this into the reflection and transmission coefficients gives

$$R_s(\theta_i) = \frac{\eta_2 \cos\theta_i - \eta_1 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}{\eta_2 \cos\theta_i + \eta_1 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}} \qquad T_s(\theta_i) = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}.$$

Theorem 2.2 (Fresnel equations for s- and p-polarized waves).

For s-polarized waves (the electric field is parallel to the interface), the reflection and transmission coefficients in terms of θ_i are

$$R_s = \frac{\eta_2 \cos\theta_i - \eta_1 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}{\eta_2 \cos\theta_i + \eta_1 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}} \qquad T_s = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}.$$

For p-polarized waves (the magnetic field is parallel to the interface), the reflection and transmission coefficients in terms of θ_i are

$$R_p = \frac{\eta_1 \cos\theta_i - \eta_2 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}{\eta_1 \cos\theta_i + \eta_2 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}} \qquad T_p = \frac{2\eta_2 \cos\theta_i}{\eta_1 \cos\theta_i + \eta_2 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}.$$

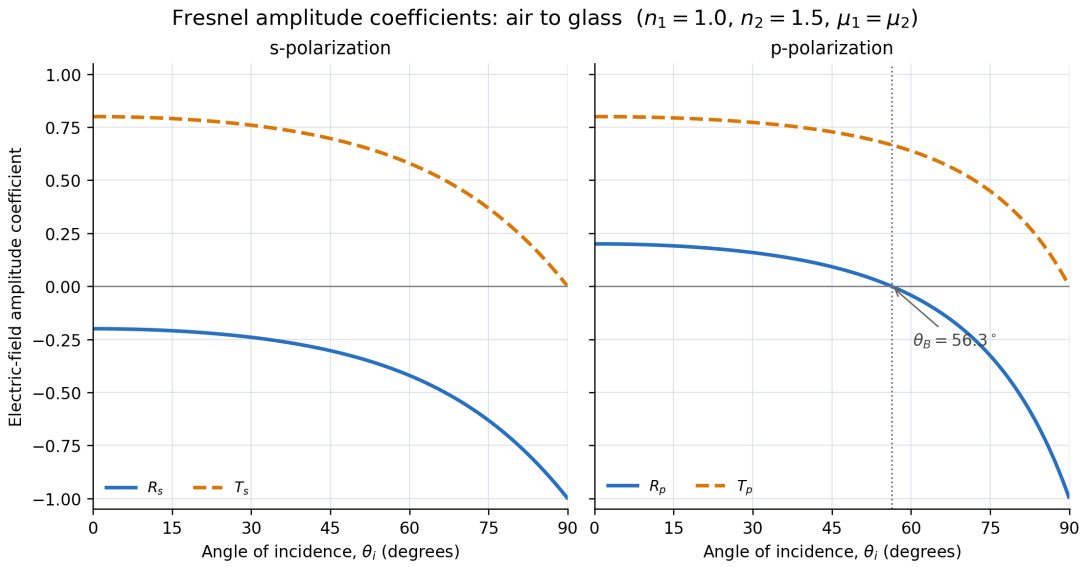


Figure 4: The fresnel coefficients for glass as a function of the incident angle. Note that for the p-polarised wave, there is a point where the reflected wave vanishes. That point is the brewster angle.

Remark 2.3

The labels s and p come from the German words senkrecht (perpendicular) and parallel. The plane of incidence is the plane containing the incident wave vector and the normal to the interface; the reflected and transmitted wave vectors also lie in this plane.
For s-polarized waves, the electric field is perpendicular to the plane of incidence. For p-polarized waves, the electric field is parallel to the plane of incidence.

2.3 The brewster angle

The Brewster angle is the angle at which the p-polarized reflection coefficient vanishes. Setting the numerator of R_p to zero gives

$$\eta_1 \cos\theta_B = \eta_2 \cos\theta_t, \qquad n_1 \sin\theta_B = n_2 \sin\theta_t.$$

Eliminating θ_t gives

$$\tan^2\theta_B = \frac{(\eta_1^2 - \eta_2^2)n_2^2}{\eta_2^2(n_2^2 - n_1^2)}.$$

For non-magnetic dielectrics, where $\mu_1 = \mu_2$, this simplifies to

$$\theta_B = \arctan\left(\frac{n_2}{n_1}\right).$$

Theorem 2.4 (Brewster angle).

The Brewster angle θ_B is the angle of incidence for which $R_p = 0$. In general, for magnetic media,

$$\tan^2\theta_B = \frac{(\eta_1^2 - \eta_2^2)n_2^2}{\eta_2^2(n_2^2 - n_1^2)}.$$

For non-magnetic dielectrics, where $\mu_1 = \mu_2$, this simplifies to

$$\tan\theta_B = \frac{n_2}{n_1}.$$